



1. Basic Information

Event Name – RoboRace

Event Type – Technical

Event Duration – Start time: 2:00pm

Venue – Ground Floor

Event Coordinators –

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2. Event Description –

RoboRace is a thrilling line tracer competition designed to challenge your engineering and programming prowess. This event casts participants as robotic engineers tasked with designing, building, and programming an autonomous robot to navigate a complex racecourse.

The mission is to create a robot that can flawlessly follow a black line on a white surface, tackling a variety of challenges including sharp turns, tricky intersections, and unexpected breaks in the path. This competition is the ultimate test of your skills in integrating sensor technology, implementing precise control systems, and optimizing your code for speed and accuracy. It's not just about crossing the finish line; it's about demonstrating superior design and intelligent navigation.

ROBOT SPECIFICATIONS:

- Each participating team must design and build a line tracer robot capable of autonomously following a visible line on the track.
- The robot must fit within specified size of 22cm x 22cm x 22cm.
- All power for the robot must be onboard; no external power sources are permitted during the race.



- Bot must be started by only one switch. However, a team may have an onboard switch.
- The autonomous bot should not separate or split into two or more units. All bots/units which are touching each other or are in the starting point will be considered as one bot.
- The Machine cannot be constructed using ready-made 'Lego kits or any ready-made mechanism. But they can make use of readymade gear assemblies. Violating this clause will lead to the disqualification of the team.
- During the run, the autonomous bot must not damage the arena in any way. It is not allowed to leave anything behind or make any marks while traversing the arena. Any bot found damaging the arena will be immediately disqualified. The final decision is at the discretion of the organizers.
- The track will consist of a visible line on a contrasting surface, featuring curves, turns, intersections, and possibly challenges.
- The track layout will be disclosed to participants before 3 days of the competition to allow for sensor calibration.
- The game field consists of an arena having dimensions 200 cm X 300 cm (l X b). It consists of the following: 1. The arena is composed of random paths made up of black Vinyl strips. 2. All the distances are shown in Map figure 3. The width of all black stripes will be 1.5mm-20mm. 4. The figure below shows the sample arena. The actual arena at the competition will consist of alterations in the path. 5. A black box of 200mm x 200 mm is present at the end zone of the arena to indicate the end position. Note: The dimensions of the arena will be accurate to within 5% or 20 mm, whichever is less.

GAMEPLAY:

- Participants are allowed a specified calibration period before the race to adjust their robots' sensors for optimal line detection.
- Each robot will start from a designated starting point on the track.
- The goal is for the robot to complete the course in the shortest time possible by accurately following the line.
- Penalties may be incurred for actions such as straying off the line, interfering with other robots, or violating size and weight limits.



- The official race time starts when the robot begins moving from the starting point and stops when it successfully completes the course.

3. Participation Guidelines

- **Eligibility** – The event is open to our college students.
- **Team Size** – A minimum of **1** and a maximum of **4** members per team.
- **Registration Process** – Registration will be conducted through both online (Google Forms) and offline (on-desk) modes. The registration fee is **100 + 400(Deposit)**.
- **Confirmation & Communication** – All registered teams will receive confirmation and subsequent event updates via a dedicated WhatsApp group.

4. Rules & Regulations

- **General Code of Conduct:** All participants must maintain professional decorum. Any form of malpractice or unsportsmanlike conduct will lead to immediate disqualification.
- **Robot Specifications:**
 - The robot must be **fully autonomous**. No wired or wireless control is permitted during the race.
 - The dimensions of the robot must not exceed **20cm x 20cm x 15cm** (L×W×H).
 - The power source must be on-board and not exceed **12V DC**.
 - Participants must bring their own programmable robots. No kits will be provided on the spot.
- **Race Rules:**
 - If a robot leaves the track, the team is allowed to place it back on the line. Each hand touch will incur a time penalty.
 - The robot must start from the designated starting point and will be timed until it crosses the finish line.



- **Disqualification Criteria:** Any team found tampering with another team's robot, violating the robot specifications, or using more than the allowed hand touches will be disqualified.

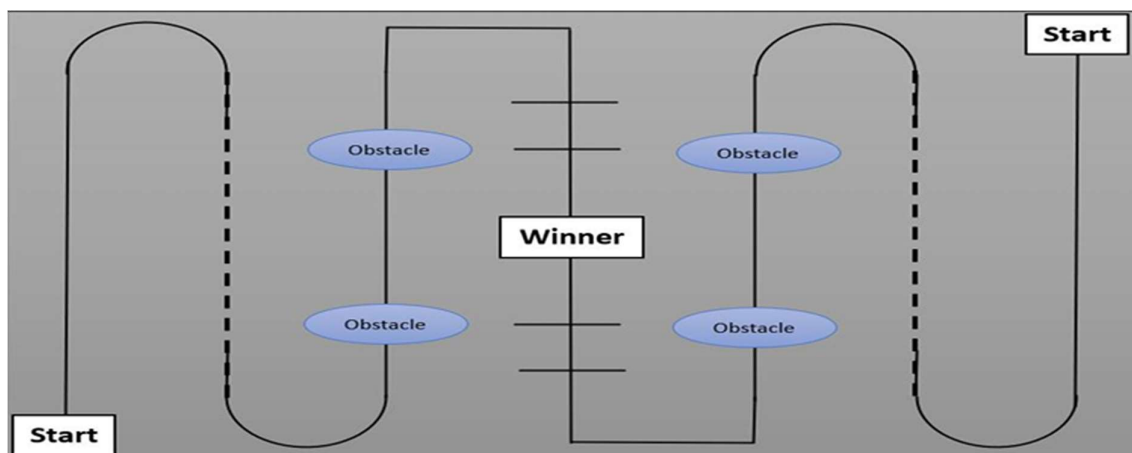
5. Event Structure & Flow

- **Number of Rounds:** The event will be conducted as a single-elimination knockout tournament. The total number of rounds will depend on the number of participating teams.
- **Round Descriptions** – Teams will be randomly paired to compete in head-to-head races. In each match, the robot that successfully completes the course in the fastest time (including penalties) will be declared the winner and will advance to the next round. The losing team will be eliminated from the competition. This process will continue through subsequent rounds (e.g., Quarter-Finals, Semi-Finals) until a final match determines the champion.
- **Flow of Activities** – Step-by-step timeline

TRACK MAP:

Note: - Map may consider as example It may change also.

Top View:



Checkpoints will be given in between start and end point, each checkpoint carries Points



6. Judging & Evaluation

- **Judging Criteria:** The event is judged on **Speed** and **Accuracy**. The primary factor is the time taken to complete the course in the final round.
- **Scoring Pattern:**
 - Performance is measured by the total time taken to traverse the track.
 - A **5-second penalty** will be added to the final time for each "hand touch" used.
 - The team with the lowest final time will be declared the winner.
- **Tie-Breaker Rules:** In the event of a tie within a single match:
 - The team that used fewer hand touches will be declared the winner.
 - If both teams have identical times and penalty counts, the match will be re-run to determine a winner.
 - If neither robot finishes the course, the one that traveled the farthest distance along the correct path will win the match.

7. List of Topics/Tasks/Challenges –

The challenge is centered around navigating a custom-built race track. The track will be a **black line of approximately 2.5 cm width** on a white surface.

Participants' robots will need to successfully navigate a course featuring:

- **Sharp Turns:** Including multiple 90-degree turns and acute-angle corners.
- **Intersections:** 'Plus' or 'T' shaped intersections where the robot must continue along the pre-defined correct path.
- **Line Breaks:** Small gaps in the track that the robot must cross using momentum or predictive logic.
- **Curved Paths:** S-curves and arcs designed to test the robot's sensor calibration and motor control.



GAME RULES:

1. Teams will be given 2 minutes for calibration. If any team is found to alter its code alter depositing its bots, then it will be immediately disqualified from the competition. They are, however, allowed to make any other hardware changes.
2. Only one autonomous bot per team is allowed.
3. When the bot starts, Team member is not allowed to touch the bot or enter the arena. Note: Only one member is allowed to touch and put bot again on nearest crossed checkpoint in-case if bot missed or unfollowed specific path alter crossing checkpoints.
4. At the start of the task, the bot will be placed at the starting point. Only 1 team member is allowed to be near the game field while starting the bot.
5. Run will start only when organizers give the signal. 6. The starting procedure of the bot should be simple and should not involve giving the bot any manual force or impulse in any direction.

GENERAL RULES:

1. Participants are not allowed to keep anything inside the arena other than the bot.
2. The time measured by the organizers will be final and will be used for scoring the teams.
3. Time measured by any contestant by any other means is not acceptable for scoring.
4. In case of any disputes/discrepancies, the organizers decision will be final and binding.
5. The organizers reserve the right to change any or all of the above rules as they deem fit. Change in rules, if any, will be highlighted on the website and notified to the registered teams.
6. Only one team is allowed to be present during the run, other teams will have to stay outside the hall. No team is allowed to take photographs or record their run.

SCORING AND WINNING CRITERIA: The winning robot is determined by the fastest completion time to reach end point. In case of ties, the robot with the highest accuracy in following the line may be considered.

